Lio Sam Algorithm Mapping

This algorithm has been successfully compiled and run on the Orin-Nano motherboard. Orin-Nano motherboard system environment: **Ubuntu22.04+ROS2-Humble**.

Startup Command

Open Terminal 1 and enter the following command to start the Lio-Sam mapping program,

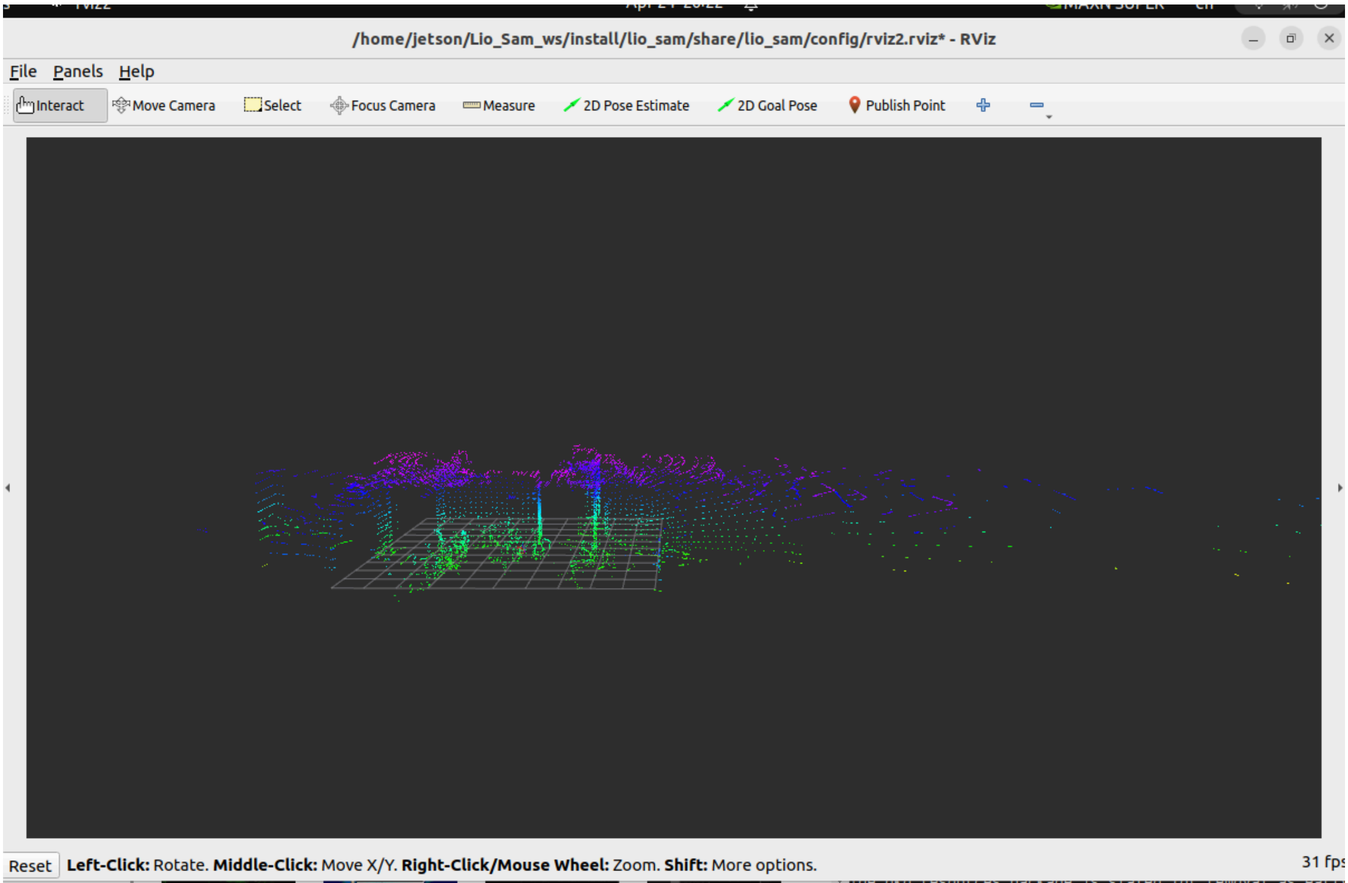
```
ros2 launch lio_sam run.launch.py
```

After the program starts, an rviz will open to display the mapping process,

Open Terminal 2 and enter the following command to start the 32-line lidar,

```
ros2 launch vanjee_lidar_sdk start.py
```

Then, slowly move the lidar. The point cloud map in rviz will be gradually generated, as shown in the following figure,



Saving the Map

Open Terminal 3 and enter the following command to save the map,

```
ros2 service call /lio_sam/save_map lio_sam/srv/SaveMap "{resolution: 0.5 , destination: /Downloads/LioSam}"
```

Here, resolution represents the sampling rate, and destination represents the folder to save. Given /Downloads/LioSam here, it means the saved point cloud map will be placed in the ~/Downloads/LioSam directory. The code adds a ~.

Then, we can see in the ~/Dowloads/LioSam directory that 5 .pcd files have been generated. Among them, GlobalMap.pcd is the global point cloud map. You can use the following command in the graphical interface to view the global point cloud map,

```
cd ~/Dowloads/LioSam # Modify the path here according to the actual save address
pcl_viewer GlobalMap.pcd
```

The parameter table is as follows,

```
/**:
  ros__parameters:

    # Topics
    pointCloudTopic: "/rslidar_points"                  # Point cloud data
    imuTopic: "/vanjee_lidar_imu_packets"                       # IMU data
    odomTopic: "odometry/imu"                   # IMU pre-preintegration odometry, same
frequency as IMU
    gpsTopic: "odometry/gpsz"                   # GPS odometry topic from navsat, see
module_navsat.launch file

    # Frames
    lidarFrame: "vanjee_lidar"
    baselinkFrame: "base_link"
    odometryFrame: "odom"
    mapFrame: "map"

    # GPS Settings
    useImuHeadingInitialization: false          # if using GPS data, set to "true"
    useGpsElevation: false                      # if GPS elevation is bad, set to "false"
    gpsCovThreshold: 2.0                        # m^2, threshold for using GPS data
    poseCovThreshold: 25.0                      # m^2, threshold for using GPS data

    # Export settings
    savePCD: true                              # https://github.com/TixiaoShan/LIO-
SAM/issues/3

    # Sensor Settings
    sensor: velodyne                              # lidar sensor type, either 'velodyne',
'ouster' or 'livox'
    N_SCAN: 32                                  # number of lidar channels (i.e.,
Velodyne/Ouster: 16, 32, 64, 128, Livox Horizon: 6)
    Horizon_SCAN: 2000                           # lidar horizontal resolution
(Velodyne:1800, Ouster:512,1024,2048, Livox Horizon: 4000)
    downsampleRate: 1                           # default: 1. Downsample your data if too
many
    # points. i.e., 16 = 64 / 4, 16 = 16 / 1
    lidarMinRange: 0.3                          # default: 1.0, minimum lidar range to be
used
```

```yaml
    lidarMaxRange: 40.0                       # default: 1000.0, maximum lidar range to be
used
    maxFeatureNum: 300


    # IMU Settings
    imuAccNoise: 3.9939570888238808e-03
    imuGyrNoise: 1.5636343949698187e-03
    imuAccBiasN: 6.4356659353532566e-05
    imuGyrBiasN: 3.5640318696367613e-05

    imuGravity: 9.80511
    imuRPYWeight: 0.01

    maxVel: 50.0
    maxIter: 4
    maxNoise: 0.01
    accGating: 0.5
    gyrGating: 0.5

    extrinsicTrans:  [ 0.0,  0.0,  0.0 ]
    extrinsicRot:    [1.0,  0.0,  0.0,
                      0.0,  1.0,  0.0,
                      0.0,  0.0, 1.0 ]
    extrinsicRPY: [1.0, 0.0, 0.0,
                   0.0, 1.0, 0.0,
                   0.0, 0.0, 1.0]

    # LOAM feature threshold
    edgeThreshold: 0.1
    surfThreshold: 0.1
    edgeFeatureMinValidNum: 5
    surfFeatureMinValidNum: 30

    # voxel filter paprams
    odometrySurfLeafSize: 0.2                    # default: 0.4 - outdoor, 0.2 - indoor
    mappingCornerLeafSize: 0.1                   # default: 0.2 - outdoor, 0.1 - indoor
    mappingSurfLeafSize: 0.2                     # default: 0.4 - outdoor, 0.2 - indoor

    # robot motion constraint (in case you are using a 2D robot)
    z_tollerance: 1000.0                         # meters
    rotation_tollerance: 1000.0                  # radians

    # CPU Params
    numberOfCores: 3                             # number of cores for mapping optimization
    mappingProcessInterval: 0.                 # seconds, regulate mapping frequency

    # Surrounding map
    surroundingkeyframeAddingDistThreshold: 1.0   # meters, regulate keyframe adding
threshold
    surroundingkeyframeAddingAngleThreshold: 0.2  # radians, regulate keyframe adding
threshold
```

```
    surroundingKeyframeDensity: 2.0                # meters, downsample surrounding keyframe
poses
    surroundingKeyframeSearchRadius: 50.0          # meters, within n meters scan-to-map
optimization
    # (when loop closure disabled)

    # Loop closure
    loopClosureEnableFlag: false
    loopClosureFrequency: 1.0                      # Hz, regulate loop closure constraint add
frequency
    surroundingKeyframeSize: 25                    # submap size (when loop closure enabled)
    historyKeyframeSearchRadius: 15.0              # meters, key frame that is within n
meters from
    # current pose will be considerd for loop closure
    historyKeyframeSearchTimeDiff: 30.0            # seconds, key frame that is n seconds
older will be
    # considered for loop closure
    historyKeyframeSearchNum: 25                   # number of hostory key frames will be
fused into a
    # submap for loop closure
    historyKeyframeFitnessScore: 0.3               # icp threshold, the smaller the better
alignment

    # Visualization
    globalMapVisualizationSearchRadius: 1000.0     # meters, global map visualization radius
    globalMapVisualizationPoseDensity: 10.0        # meters, global map visualization
keyframe density
    globalMapVisualizationLeafSize: 1.0            # meters, global map visualization cloud
density
```

Debug and modify according to actual situations.